

The Control Chart

An old friend, the run chart, with three new lines that remove the guesswork about signal versus noise.

The Three Lines

In the middle is the center line, the average of your process. Above and below are the upper and lower control limits. Crucially, these are not targets and not the same as specification limits or goals. They are calculated from your own process data, usually three standard deviations out, and the tools do the math for you.

Inside Or Outside

If a point falls inside the limits, that is common cause variation, ordinary noise, and the correct response is to do nothing. If a point falls outside, that is a special cause signal, something unusual happened, and you investigate. The chart does your signal versus noise judgment automatically, every time a point lands.

The Guardian

After you improve and standardize, you keep plotting new data. While points stay inside the limits your gains are holding. A point poking outside is an early warning of drift while it is still small and fixable. Watch for other patterns too, like a long run of points on one side of the center line.

Remember this

A control chart is a smoke detector for your process, but knowing what changed stays human.